



LaSalle County Station
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10 CFR 50.73

RA17-037

April 18, 2017

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

LaSalle County Station, Unit 1
Renewed Facility Operating License No. NPF-11
NRC Docket No. 50-373

Subject: Licensee Event Report 2017-005-00, Manual Reactor Scram Resulting
From Feedwater Regulating Valve Failure Causing High Reactor Water
Level

In accordance with 10 CFR 50.73(a)(2)(iv)(A), Exelon Generation Company, LLC
(EGC) is submitting Licensee Event Report (LER) Number 2017-005-00 for LaSalle
County Station, Unit 1.

There are no regulatory commitments in this letter. Should you have any questions
concerning this report, please contact Mr. Guy V. Ford, Jr., Regulatory Assurance
Manager, at (815) 415-2800.

Respectfully,

A handwritten signature in black ink, appearing to read "William J. Trafton".

William J. Trafton
Site Vice President
LaSalle County Station

Enclosure: Licensee Event Report

cc: Regional Administrator – NRC Region III
NRC Senior Resident Inspector – LaSalle County Station

**LICENSEE EVENT REPORT (LER)**

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME

LaSalle County Station, Unit 1

2. DOCKET NUMBER

05000373

3. PAGE

1 OF 3

4. TITLE

Manual Reactor Scram Resulting From Feedwater Regulating Valve Failure Causing High Reactor Water Level

5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED	
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME	DOCKET NUMBER
02	17	2017	2017	005	00	04	18	2017	NA	NA
9. OPERATING MODE			11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: (Check all that apply)							
1			☐ 20.2201(b)	☐ 20.2203(a)(3)(i)		☐ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)		
			☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)		☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)		
			☐ 20.2203(a)(1)	☐ 20.2203(a)(4)		☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)		
			☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)		☒ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)		
10. POWER LEVEL 030			☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)		☐ 50.73(a)(2)(v)(A)		☐ 73.71(a)(4)		
			☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)		☐ 50.73(a)(2)(v)(B)		☐ 73.71(a)(5)		
			☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)		☐ 50.73(a)(2)(v)(C)		☐ 73.77(a)(1)		
			☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)		☐ 50.73(a)(2)(v)(D)		☐ 73.77(a)(2)(i)		
			☐ 20.2203(a)(2)(vi)	☐ 50.73(a)(2)(i)(B)		☐ 50.73(a)(2)(vii)		☐ 73.77(a)(2)(ii)		
			☐ 50.73(a)(2)(l)(C)		☐ OTHER		Specify in Abstract below or in NRC Form 366A			

12. LICENSEE CONTACT FOR THIS LER

LICENSEE CONTACT

John Kowalski, Engineering Director

TELEPHONE NUMBER (Include Area Code)

(815) 415-3800

13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT

CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX
B	SJ	FCV	C600	Y	NA	NA	NA	NA	NA

14. SUPPLEMENTAL REPORT EXPECTED☐ YES (If yes, complete 15. EXPECTED SUBMISSION DATE) ☒ NO**15. EXPECTED SUBMISSION DATE**

MONTH	DAY	YEAR
NA	NA	NA

ABSTRACT (Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)

On February 17, 2017, at 2353 CST, during Unit 1 power ascension from a previous forced shutdown, operators inserted a manual scram as a result of a high reactor water level condition caused by a rapid change in feedwater flow. The high reactor water level trip occurred due to a failure of the feedwater regulating valve (FRV) 1FW005 positioner arm, which caused the regulating valve to be driven to the full open position. This resulted in a rapid increase of reactor water level that required operators to perform a manual reactor scram. The plant was placed in a stable condition, with no complications in achieving shutdown.

This event is reportable in accordance with 10 CFR 50.73(a)(2)(iv)(A) as an event or condition that resulted in manual or automatic actuation of the reactor protection system (RPS). The safety significance of this condition was minimal, as plant equipment responded as expected for the event. The feedback drive arm and new positioner were installed on the Unit 1 FRV, which allowed the unit to restart following repairs. The FRV positioner feedback assembly had been installed in 2002, during a modification to upgrade of the air-operated valve assembly. This design has proven not to be robust enough for the FRV application.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
LaSalle County Station, Unit 1	05000373	2017	- 005	- 00

NARRATIVE**PLANT AND SYSTEM IDENTIFICATION**

LaSalle County Station Unit 1 is a General Electric Boiling Water Reactor with 3546 Megawatts Thermal Rated Core Power.

The Feedwater (FW) system provides a source of processed water to the reactor pressure vessel (RPV) to maintain the desired water level during normal plant operating conditions. The affected component was the Feedwater Regulating Valve (FRV), which is used to control the flow of FW to the RPV for water level control when the motor-driven reactor feed pump is used.

CONDITION PRIOR TO EVENT

Unit(s): 1	Event Date: February 17, 2017	Event Time: 2353 CST
Reactor Mode(s): 1	Mode(s) Name: Power Operation	Power Level: 30 percent

DESCRIPTION OF EVENT

On February 17, 2017, Unit 1 was in the process of power ascension at approximately 30 percent power from a previous forced shutdown. At 23:53 CST, the Main Control Room received a FW control RPV high level alarm. Operators quickly diagnosed an abrupt change in reactor water level and that level was rapidly increasing. The operators entered the abnormal procedures in parallel with setting manual scram criteria of greater than 50 inches reactor water level and not in control, as directed by station procedures. The manual scram was initiated as level was approaching the turbine trip set point of 55.5 inches in order to terminate the transient. The scram was not complicated, and all equipment performed as designed. The plant was placed in a stable condition, with reactor pressure maintained by the turbine bypass valves. Reactor water level was controlled with FW through the low-flow regulating valve. All control rods fully inserted, and decay heat was removed via steam to the main condenser using bypass valves.

CAUSE OF EVENT

The high reactor water level trip occurred due to a failure of the FW regulating valve (FRV) positioner arm, which caused the regulating valve to be driven to the full open position and resulted in a rapid increase in RPV level. Field observations at FRV 1FW005 revealed that the valve positioner feedback drive arm had broken cleanly into two pieces at the junction between the arm and shaft. The feedback drive arm extension was also found to be severely bent. The FRV positioner feedback assembly was a standard vendor design when installed as part of the hydraulics for the upgrade of the air-operated valve (AOV) performed in 2002. This design of the valve position feedback assembly has proven not to be robust enough for the FRV application. Contributing to this event were FRV cycling during operations caused by the inherent dead-band of the valve positioner and the positioner feedback assembly.

REPORTABILITY AND SAFETY ANALYSIS

The event was reportable to the NRC in accordance with 10 CFR 50.72(b)(2)(iv)(B) for any event or condition that results in actuation of the reactor protection system (RPS) when the reactor is critical. An Emergency Notification System (ENS) report number 52564 was made on February 18, 2017.

This event is also reportable in accordance with 10 CFR 50.73(a)(2)(iv)(A) as an event or condition that resulted in manual or automatic actuation of the reactor protection system (RPS). The safety significance of this condition was minimal, as plant equipment responded as expected for the event; therefore, there were no impacts to plant safety or design margins.

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NARRATIVE**CORRECTIVE ACTIONS**

A replacement feedback positioner arm and a new positioner were installed on the Unit 1 FRV. A calibration of the FRV positioner arm and a FlowScan diagnostic test were performed to verify proper operation. Additional corrective actions are to evaluate the design of the Motor Driven Reactor Feed Pump (MDRFP) FRV positioner and its feedback assembly for potential improvements that could make the feedback arm more robust and have less dead-band to reduce the valve cycling.

PREVIOUS OCCURRENCES

A review of past events in the last three years identified no previous occurrences that related to FW valve or reactor water level control events. Examples of reactor scrams included Unit 2 LER 374-2014-001-00 (Reactor Scram Due to Main Steam Isolation Valve Stem-Disk Separation) and Unit 2 LER 374-2017-001-00 (Manual Reactor Scram due to Turbine-Generator Run-Back Caused by Stem-Disk Separation in Stator Water Cooling Heat Exchanger Inlet Valve), and Unit 1 LER 373-2017-003-00 (Automatic Reactor Scram due to Main Generator Trip on Differential Current During Back-Feed Operations). However, none of these were applicable to the subject event. Searches were also performed of the station's corrective action program and INPO operating experience databases, which did not identify information or a similar occurrence to this event.

COMPONENT FAILURE DATA

Manufacturer: Control Components International (C600)
Device (Model): Air-Operated Valve Regulator Assembly (922401010, Self-Drain Element)
Component ID: 1FW005, Motor Driven Reactor Feed Regulating Valve "C"